Understanding some physics behind it:

First Law of Thermodynamics (conservation of energy):

Q = m*C* delta T

{Q = heat dissipation rate (W)
m = mass flow rate
C = specific heat capacity
Delta T = temperature difference between T inlet and T outlet}

Q ∝ sqrt(Delta P) * Delta T

We need sensors to track *both* Delta P (pressure drop) and Delta T (temperature difference) for an accurate result.

Why isn't temperature measurement enough to monitor battery cooling? Why do we need pressure sensors as well?

Temperature sensors have an inherent physical lag. For example, if an air duct gets blocked by debris, the pressure drop we measure drops to 0 instantly. A pressure sensor would notify us of cooling failure in milliseconds. This is great because the car can respond before the batteries have time to absorb the trapped heat.

Additionally, measuring the pressure drop lets us identify the root cause of the failure. For example, is the temperature spiking because of a blocked air duct, a dying fan, or is there an internal electrical fault?